SOORAJ SUDHAKARAN

Robotics & Automation Engineer | M.Eng Mechatronics
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PROFESSIONAL SUMMARY

Master's student in Mechatronics and Cyber-Physical Systems seeking full-time engineering roles in robotics, control systems, and industrial automation. Hands-on experience in ROS-based autonomy, simulation-driven design, and embedded/real-time concepts. Strong focus on safe robotic motion, sensor integration, and practical implementation from simulation to real hardware.

CORE SKILLS

Robotics: ROS/ROS2, navigation, SLAM, motion control, trajectory planning, robot programming, sensor integration, safety-aware motion planning
Programming: C++ (OOP), Python, C#, ROS, MATLAB, Simulink
XR / Simulation Development: Unity, Virtual Reality (VR), Augmented Reality (AR), Extended Reality (XR), Human-Machine Interaction
Computer Vision & Graphics: OpenCV, Camera-based tracking, Fusion 360
Sensor Technologies: LiDAR, Radar, Sensor Fusion
Simulation and CAD: SOLIDWORKS, AutoCAD, ANSYS Workbench, Blender
Additive Manufacturing: DfAM, Topology Optimization, FDM/SLA
Tools & Systems: GitHub, Visual Studio Code, Windows, Ubuntu
Functional Safety: ISO 26262 (concepts and lifecycle)

WORK EXPERIENCE

MASTER'S THESIS | INDUSTRIAL AUTOMATION AND ROBOTICS | Full time

KEBA Group, Germany (Onsite) 09/2025 – Present

- Developed an automated robot-programming framework for sheet-metal bending operations, generating executable programs for the KEBA robotic controller and improving programming efficiency in variant-rich production.
- Implemented Python-based sequencing, re-grip strategies, and collision-avoidance logic to ensure safe, reliable robotic manipulation across diverse workpieces.
- Integrated simulation and real-cell constraints through ROS WebSocket connectivity, enabling robust real-time communication and seamless deployment to the industrial robotic controller.

Skills: Python, ROS, Industrial Robotics, Control Logic, Robot Programming, Simulation, WebSocket Communication

NON-DESTRUCTIVE TESTING TECHNICIAN | Full time

02/2022 – 09/2022
United Engineering and Construction Co., Cochin, Kerala, India (Onsite)

- Applied advanced Non-Destructive Testing (NDT) techniques to assess material integrity and ensure structural safety while conducting inspections and documenting quality evaluations in accordance with engineering and site standards.

Skills: Non-Destructive Testing, Ultrasonic Testing, Magnetic Particle Testing, Radiographic Testing

EDUCATION

MASTER OF ENGINEERING, MECHATRONIC AND CYBER-PHYSICAL SYSTEMS

03/2023 – Present
Technische Hochschule Deggendorf (THD), Germany Current Grade: 2.3
Relevant Courses: Advanced Robotics, Autonomous Systems, Cyber-Physical Systems, Additive Manufacturing, Human-Machine Interaction

BACHELOR OF TECHNOLOGY, MECHANICAL ENGINEERING

08/2017 – 07/2021
APJ Abdul Kalam Technological University, Kerala, India Grade: 8.0/10
Relevant Courses: CAD and CAE, Material Science, Programming (C), Fluid and Thermal Engineering

PROJECTS

MASTER'S PROJECT, DESIGNED AND IMPLEMENTED AN AUTONOMOUS VACUUM ROBOT IN ROS 03/2024 – 06/2024

- Designed and implemented an autonomous vacuum robot with real-time localization using SLAM, optimizing obstacle detection and path planning for improved efficiency.
- Implemented sensor fusion with LiDAR, cameras, and obstacle detection.

Skills: ROS, Autonomous Navigation, Path Planning, Sensor Integration, SLAM, Motion Control, Embedded Systems, Obstacle Avoidance

MASTER'S PROJECT, DEVELOPED A VR WORKSHOP TO OPERATE A MACHINE USING UNITY 03/2024 – 06/2024

- Integrated real-time physics and user interactions for an authentic experience.
- Developed interactive VR-based machine operation workshop with real-time simulation and motion tracking.

Skills: Unity, Virtual Reality (VR), Augmented Reality (AR), Mixed Reality, 3D Modeling, UI/UX Design, Human Machine Interface

MASTER'S PROJECT, TOPOLOGY OPTIMISED TEMPORARY BAG SEALER USING SOLIDWORKS FOR ADDITIVE MANUFACTURING 10/2023 – 01/2024

- Applied FEA-driven topology optimization, achieving 22% weight reduction and 34% material cost savings.
- Prepared the design for 3D printing.

Skills: Topology Optimization, Finite Element Analysis (FEA), 3D Printing, Design for Manufacturing (DFM)

MASTER'S PROJECT, ACTIVE SUSPENSION SYSTEM MODELING USING MATLAB SIMULINK 10/2023 – 01/2024

- Implemented hydraulic actuators for real-time suspension control and performance optimization.
- Simulated and analysed system dynamics to refine responsiveness and efficiency.

Skills: MATLAB, Simulink, System Modeling, Dynamic Simulation

BACHELOR PROJECT, AUTOMATED MECHATRONIC ROBOT FOR ENHANCING HOSPITALITY SERVICES 07/2021

Sree Buddha College of Engineering, Kerala, India

- Programmed an Arduino-based motion control system for a hospital robot with line-tracking navigation. Integrated sensors to improve path accuracy and ensure smooth operation in dynamic environments.

Skills: Arduino Programming, Motion Control, Robotics, Sensor Integration, Autonomous Navigation

CERTIFICATIONS

Industrial thesis / internship certificate — KEBA Group (2026)
Robot Operating System, edX (2024)
Non-Destructive Testing (Level 2), United NDT Training Center (2022)
Process piping and quality control, United NDT Training Center (2022)
Internship in Mechatronics, ZINDOT Technologies (2020)
Telerobotics — VHB (2024)
AutoCAD workshop (2019)

LANGUAGES

English: Fluent (C1) | German: A2 | Malayalam: Native

PERSONAL INTERESTS

Travelling, Watching Sci-fi movies, Cooking, playing Online Games .

ACCOMPLISHMENTS

Awarded Best Bachelor Project of the College (2021) for automated mechatronic robot in hospitality services; served as Secretary of the Mechanical Engineering Department Association (2018–2021).

Sooraj Sudhakaran